

Standard Operating Procedure

Work Package 11 - Genome Applications

Task 11.3 - BioGenomeApp Network

GAP ANALYSIS

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This subtask will identify the current scientific gaps that are impeding the full implementation of genetic and genomic applications in biodiversity assessments. A number of short, medium, and long-term needs will be identified to enable full integration of genetic approaches into the development of application toolkits for biodiversity research. It will catalogue the specific added value of reference genomes to application development, as well as best practices for population sampling, alternative data production strategies, and available analytical toolkits, as well as describe how the information gathered from genetic and genomic analyses can be used to inform practitioners and relevant actors in three major application areas: conservation, bioeconomy, and disease.

The gap analysis will consist of four major steps: **1. Assess the current situation (identifying the gap), 2. Characterise and analyse the gap, 3. Propose mitigation measures, and 4. Describe future goals.** These will result in a report, which has been indicated as a Task milestone, followed by publication(s) in peer review journals detailing the findings and conclusions.

Taking advantage of the diversity of case studies included in the Work Package 11 (WP11), we will also conduct a gap analysis. The analysis will consider the stakeholders needs identified for each case study in the tasks T11.1 and T11.2 and the current gaps that hinder development of biodiversity applications.

This **Standard Operating Procedure** intends to establish a common strategy for the development of T11.3 Gap Analysis and a common procedure for literature mining, metadata analysis, and information gathering within the scope of the three following perspectives: **From Basic to Applied Science, Inequalities and Communication gaps between “interested parties” (from scientist to stakeholders), and From Applied Science to Stakeholders engagement.**

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1. Aims and Strategy for the Gap analysis

To conduct a thorough examination of the existing gap hindering the use of genetics and genomics in applied science, our analysis will employ a tripartite approach encompassing three distinct perspectives: i) From Basic to Applied Science, ii) Inequalities and Communication gaps between relevant actors (from scientist to stakeholders), and iii) From Applied Science to Stakeholders engagement. Each perspective will focus on specific factors contributing to the gap and offer complementary outcomes and solutions to facilitate the integration of genomics in practical applications. Accordingly, we intend to carry out the following tasks:

i) From Basic to Applied Science

- Investigate the application of genetics and genomics analyses, specifically focusing on their integration in conservation studies through a comprehensive literature review.
- Determine the factors preventing the widespread application of genetics and genomics in conservation, bioeconomy and disease studies by identifying the barriers and challenges encountered in utilising reference genomes.
- Assess the importance of incorporating reference genomes in genomics to develop application toolkits, aiming to understand their role in facilitating effective conservation and management strategies.
- Identify appropriate metrics to evaluate the current status of different species, considering the variations in suitable metrics across different taxonomic groups and addressing their specific management and conservation needs.
- Identify fields of research that require further development, which are relevant to the development of applications in biodiversity conservation, disease studies, and bioeconomy.
- Investigate the barriers hindering the routine use of genetics and genomics in practical applications, such as a lack of expertise (complexity), lack of appropriate infrastructure, lack of data (e.g. availability of reference genomes) and availability of funds, and determine potential strategies to overcome these barriers.
- Identify ways to make genetic and genomic analysis accessible to scientists and practitioners, in order to bridge the gap between genomic research and its practical applications in conservation and management.
- Recognize the importance of a comprehensive approach that encompasses the entire spectrum, from basic to applied science, and from applied science to stakeholders, in order to fully realise the potential of genomics in applied toolkits for biodiversity research.

ii) Inequalities and communication gaps

- Assessing policy and public awareness regarding the significance of genomics in biodiversity management:
 - a) Investigate the level of policy and public understanding and recognition of the role that genetics plays in biodiversity conservation and management.
 - b) Analyse the extent to which policymakers, relevant stakeholders and public/citizens are aware of the potential benefits and implications of genetics in addressing biodiversity challenges.
- Examining funding challenges and disparities in European countries:

- a) Identify and evaluate the existing inequalities and discrepancies in funding opportunities for genetic research in different European countries.
 - b) Investigate the impact of these funding disparities on the advancement of genetic applications in biodiversity conservation and management.
 - c) Assess potential strategies and mechanisms to promote more equitable funding opportunities and support for genetic research across European nations.
- Emphasising the importance of transnational cooperation in biodiversity research:
 - a) Recognize the significance of collaborative efforts and transnational cooperation in advancing genetic research for biodiversity conservation.
 - b) Highlight the benefits of sharing resources, expertise, and data across borders to facilitate more comprehensive and effective genetic studies.
 - c) Investigate existing collaborative frameworks and initiatives in place to promote transnational cooperation in biodiversity genetic research.
- Evaluating the role of infrastructures in facilitating genomics research:
 - a) Assess the availability and accessibility of essential infrastructures, such as high-throughput sequencing facilities, data storage, and computational resources, for conducting genomics research for biodiversity conservation and management.
 - b) Analyse how the adequacy and quality of infrastructures impact the progress and utilisation of genetics in biodiversity conservation.
 - c) Identify potential gaps or limitations in existing infrastructures and propose strategies to enhance their accessibility and capacity to support genetic research in biodiversity management.

iii) From Applied Science to Stakeholders

- Definition of stakeholders:
 - a) Provide a comprehensive definition of stakeholders relevant to the field of genetics and genomics in biodiversity management and conservation.
 - b) Identify key actors involved in conservation, stock management, public health and related areas.
 - c) Establish a comprehensive framework that systematically identifies, assesses, and engages stakeholders in the realm of genetics research utilising case studies in WP11 as a pilot project:
 - a) Incorporate case studies integrated within Work Package 11 (WP11) as a pilot initiative to facilitate the development of the gap analysis.
 - b) Explore the specificities of the case studies to gain insights into the challenges and opportunities in applying genetics in biodiversity monitoring.
- Identification of stakeholder necessities:
 - a) Engage with relevant/key stakeholders, including conservation practitioners, stock managers, and policy makers, to identify their specific needs and requirements.
 - b) Establish effective communication channels to bridge the gap between research findings and their practical application, thereby improving the transfer of knowledge and understanding among scientists and stakeholders.
- Understanding the knowledge and awareness of gaps among stakeholders (conservation practitioners, policy makers, citizen scientists, etc.):

- a) Assess the involvement of the stakeholders (conservation practitioners, policy makers, funding agencies, citizen scientists and others) in applied genomics.
 - b) Highlighting the role of stakeholder in the different phases of the applied genetic projects.
 - c) Highlighting the usefulness of genomics in biodiversity conservation and management.
 - d) Emphasise the need for enhanced awareness and education in genetics and genomics.
- Recognizing the diverse levels of stakeholder engagement in genetic applications across different taxonomic groups:
 - a) Identify the taxonomic groups that have a stronger emphasis on applied genetics and also exhibit more active stakeholder involvement.
 - b) Consider the variations in genomics knowledge, available resources, and technological advancements across taxonomic groups when addressing the specific gaps and requirements of each group.

By pursuing these objectives within each respective perspective, we aim to shed light on the identified gaps and contribute to the advancement of genomics in applied science, particularly in the field of biodiversity.

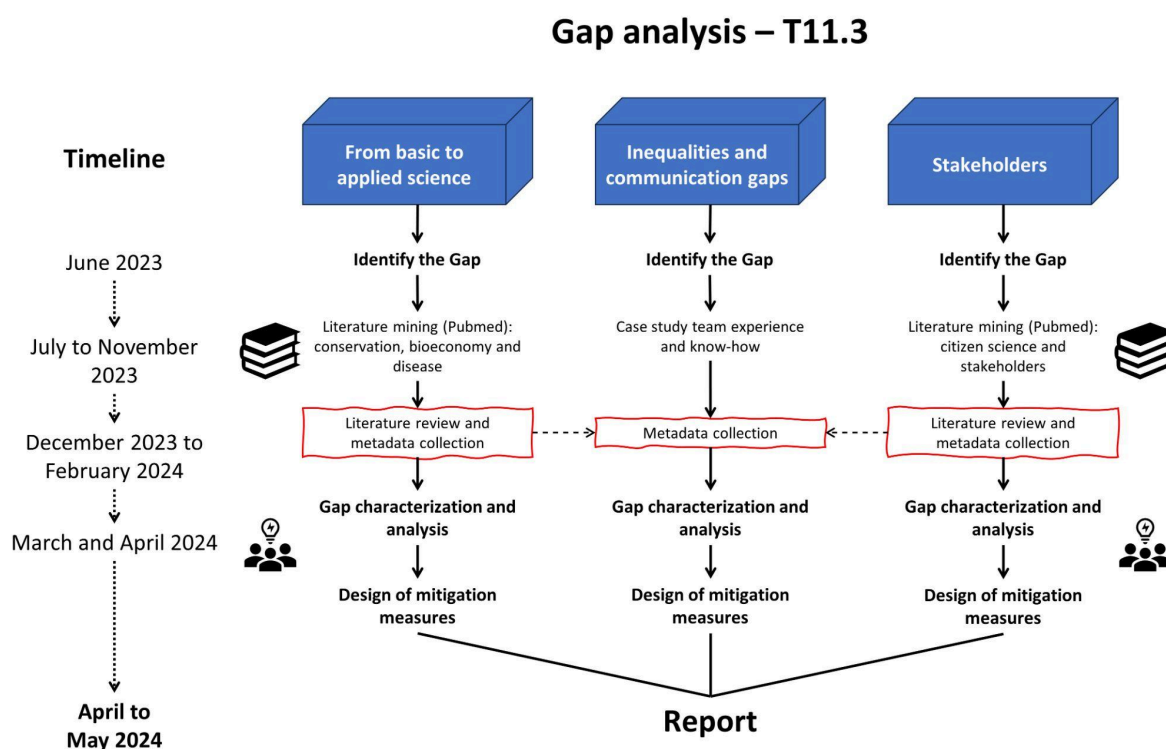


Figure 1. Gap Analysis organisational structure and workflow.

2. Definition of genomic application

An application toolkit for genomics in the field of conservation, bioeconomy, and disease is a comprehensive set of tools, knowledge, resources, and protocols specifically tailored for the application of genomic strategies and methods by practitioners and non-scientists in distinct domains:

Conservation: In this context, the toolkit encompasses tools and methodologies that leverage genetics and genomics to address issues related to the preservation and protection of biodiversity. This may involve DNA sequencing and analysis to study endangered species, track wildlife populations, or identify genetic variations critical for conservation efforts.

Bioeconomy: This part of the toolkit is geared towards utilising genetic and genomics to promote sustainable and economically viable practices in various industries, such as agriculture, fishing and hunting. Such tools are designed to address various aspects of conservation, breeding, and management to ensure the economic value of the species, while conserving their genetic diversity and long-term viability in their ecosystems.

Disease: The toolkit addresses the application in understanding, diagnosing, and combating diseases in natural populations of animals, or plants. It includes resources for genomic sequencing, bioinformatics analysis, and the identification of genetic factors contributing to disease susceptibility, which can inform veterinary treatments, plant disease management and population management. It can focus on the factors spreading the disease, e.g. vectors, or the biological populations affected by the disease.

Other. Toolkits that address other challenges in natural populations of animals or plants, for the protection of biodiversity.

An application toolkit in these fields should allow i) generating knowledge that influences immediate actions of researchers, practitioners, citizen scientists and/or policymakers in the management of the focal species, and/or ii) develop genetic tools and tests that can be directly used by these actors in monitoring biodiversity or making autonomous decisions for the management of biodiversity.

To properly identify gaps in the implementation of genomic applications as defined above, a literature review will characterise the development of genetic applications in general, to provide context to the present use of genomics and reference genomes in biodiversity applications.

3. Rules and guidelines for literature mining

3.1. Ensure the robustness and relevance of our Gap Analysis development

We have identified Pubmed (<https://pubmed.ncbi.nlm.nih.gov/advanced/>) as the suitable database for conducting our literature search.

3.2. Development of Search Strategies

To create a comprehensive search strategy, we will employ effective techniques such as the utilisation of relevant keywords and the implementation of Boolean operators (AND, OR, NOT). These strategies aim to enhance both the precision and recall of the search results. The selection of keywords will depend on the specific topic under investigation and the scope of the literature search.

i. From Basic to Applied Science

The literature mining process for this section will primarily concentrate on the research areas encompassed by the case studies integrated within Work Package 11 (WP11), namely **conservation**, **bioeconomy**, and **disease**. To ensure the selection of pertinent literature, clear inclusion and exclusion criteria will be established. These criteria will serve as guidelines for determining which articles will be considered for the gap analysis, thereby refining the focus to encompass only the most relevant literature in the field. These criteria include five fields: a) the use of reference genome; b) the field of research; c) the use of genomics or genetics in the study; d) the use or development of an application, and e) the application of the study results in biodiversity management.

- Relevant Keyword for **Conservation**:

- a) Conservation / population conservation / species conservation / biological conservation / population monitoring / population management / habitat conservation / conservation monitoring / threat / threatened / extinction
- b) Genomics / genetics / conservation genetics
- c) genetic diversity / population structure / trend / gene flow / population size / population trends
- d) species

Example:

("Conservation" OR "population conservation" OR "species conservation" OR "biological conservation" OR "population monitoring" OR "population management" OR "habitat conservation" OR "conservation monitoring" OR "threat" OR "threatened" OR "extinction") AND ("genomics" OR "genetics" OR "conservation genetics") AND ("genetic diversity" OR "population structure" OR "trend" OR "gene flow" OR "population size" OR "population trends") AND ("species") - 6235 results - October 3, 2023

- Relevant Keyword for **Bioeconomy**:

- a) Bioeconomy / economy / game species / stock management / sustainable bioeconomy / bioeconomy strategies / bioeconomy research
- b) Genomics / genetics
- c) genetic diversity / population structure / trend / gene flow / population size / population trends
- d) species

Example:

("Bioeconomy" OR "economy" OR "game species" OR "stock management" OR "sustainable bioeconomy" OR "bioeconomy strategies" OR "bioeconomy research") AND ("genomics" OR "genetics") AND ("genetic diversity" OR "population structure" OR "trend" OR "gene flow" OR "population size" OR "population trends") AND ("species") - 118 results - October 3, 2023

- Relevant Keyword for **Disease**:

- a) Disease / pathogenesis / pathogen / epidemiology / infectious diseases / disease outbreak / resilience to diseases / etiologic agent / mortality
- b) Genomics / genetics
- c) genetic diversity / population structure / trend / gene flow / population size / population trends
- d) species

Example:

("disease" OR "pathogenesis" OR "pathogen" OR "epidemiology" OR "infectious diseases" OR "disease outbreak" OR "resilience to diseases" OR "etiologic agent" OR "mortality") AND ("genomics" OR "genetics") AND ("genetic diversity" OR "population structure" OR "trend" OR "gene flow" OR "population size" OR "population trends") AND ("species") **NOT ("human" OR "patient")** - 3658 results - October 3, 2023 - The NOT ("human" OR "patient") was included in order to exclude all results related to human diseases

ii. Inequalities and communication gaps:

To conduct a comprehensive evaluation of the existing inequalities and communication gaps in research in Europe, our primary approach will involve leveraging the experience and expertise of the work group responsible for the Gap Analysis. This workgroup comprises research teams representing over 10 European countries, ensuring a diverse and extensive perspective on the subject matter. Based on the retrieved metadata from literature mining, it will be possible to identify lesser-explored fields of research and taxonomic groups. It will enable us to characterise variations in genomics implementation across Europe. These inequalities can be observed in various aspects, such as funding, resources, infrastructure, and collaborations. Some regions or countries may have limited access to research funding or advanced facilities, leading to scientific disparities. Additionally, communication gaps

between researchers in different regions hinder knowledge sharing and collaboration, further exacerbating inequalities. By harnessing the collective knowledge and insights of this multinational work group, we aim to provide a comprehensive assessment of the current state of inequalities and communication gaps and promote solutions that offer equal opportunities for research and improved communication channels to foster collaboration and knowledge exchange across Europe.

iii. From Applied Science to Stakeholders:

The perspective of **From Applied Science to Stakeholders** will also draw significant advantages from the expertise of the research groups involved in the workgroup. Similarly to the **Inequalities and Communication Gaps** perspective, the research groups' integration brings valuable experience and knowledge to address the specific challenges and opportunities related to stakeholders and citizen science in biodiversity applications. By tapping into the collective expertise of these research groups, we aim to enhance our understanding of the gaps and potential solutions in the application of genetics, genomics and reference genomes in the field, particularly in the context of stakeholder engagement and collaboration, including citizen scientists and conservation practitioners. In addition, we will conduct comprehensive literature mining to investigate the current status of translating genomics into conservation practices by mapping stakeholders mentioned in the manuscripts. We will also investigate practical application of the outcomes by relevant stakeholders. This literature-mining process will allow us to gather relevant information and insights from existing studies, publications, and research papers. This will enable us to understand the extent to which genomics is being utilised in conservation practices and how stakeholders are incorporating it into their work.

In general, stakeholders are defined as professionals from organisations, companies, or governmental authorities who have a legitimate interest in or are affected by genetic/genomic diversity monitoring. The stakeholders are categorized into different types based on their professional backgrounds and diverse interests.

- Relevant keywords for **Stakeholders**:

- a) Stakeholder / conservation organisation / government agency(ies) / Non-governmental organization / wildlife organisation / landowner / practitioners / hunter organisation / fisheries / forestry / management organisation/ policy / policy makers / citizen scientists / industry / economy / citizen science / public health
- b) Genomics / genetics
- c) genetic diversity / population structure / genetic monitoring / population size
- d) species

Example:

("Stakeholder" OR "conservation organisation" OR "government agency" OR "Non-governmental organisation" OR "wildlife organisation" OR "landowner" OR "practitioners" OR "hunter organisation" OR "fisheries" OR "forestry" OR "management organisation" OR "policy" OR "policy makers" OR "citizen scientists" OR "industry" OR "economy" OR "citizen science" OR "public health") AND ("genomics" OR "genetics") AND

("genetic diversity" OR "population structure" OR "genetic monitoring" OR "population size")
AND ("species") - 428 results - October 3 2023

Extraction of article list from pubmed. Once the search results are generated, save the list of articles in **PubMed** format that can be selected in the "**Save**" option.

4. Literature Review Process

A Stepwise Approach

Screening the search results based on the predefined inclusion and exclusion criteria is essential to identify the relevant articles for the Gap Analysis. We will employ a systematic flowchart (Fig.2) for the initial screening process - **STEP 1** - to ensure that the selected research articles align with the objectives of the gap analysis. We will start by assessing only the abstracts, methods, tables and figures of the retrieved articles to gauge their relevance. If an article appears potentially relevant, proceed to obtain and evaluate its full text for a more comprehensive assessment in **STEP 2**. Later, the relevant papers will be redistributed to allow detailed analysis by more than one reviewer - **STEP 3**.

Within the scope of the literature review, the articles to be comprehensively analysed are those that use empirical genetic/genomic data that claim to have generated results that can be applied by practitioners or non-scientists.

The detailed instructions for filling the worksheet are given in section **5 - Instructions for the Literature Review, 5.3 - Detailed instructions to fill the research article data** starting in page 15.

4.1. STEP 1 - Identification of relevant articles (see Fig. 2)

- A. Collect the country of the first affiliation for the first and last author.
- B. Check abstract, tables and figures to evaluate if empirical data was used in the research article. If not, exclude the article.
- C. Check if genetic or genomic empirical data was used to achieve the research objectives. If not, exclude the article.
- D. Check if in the abstract, introduction or methods it is claimed that a genetic or genomic toolkit was developed for biodiversity application. It should be stated that the results of the research can be applied to address biodiversity matters related with conservation, bioeconomy, disease or other aspects. If there is no such claim, exclude the article. If in doubt, include the article (the application will be reassessed in STEP 2).
- E. If the answer to B, C and D is positive, move to the STEP 2 for extraction of information.

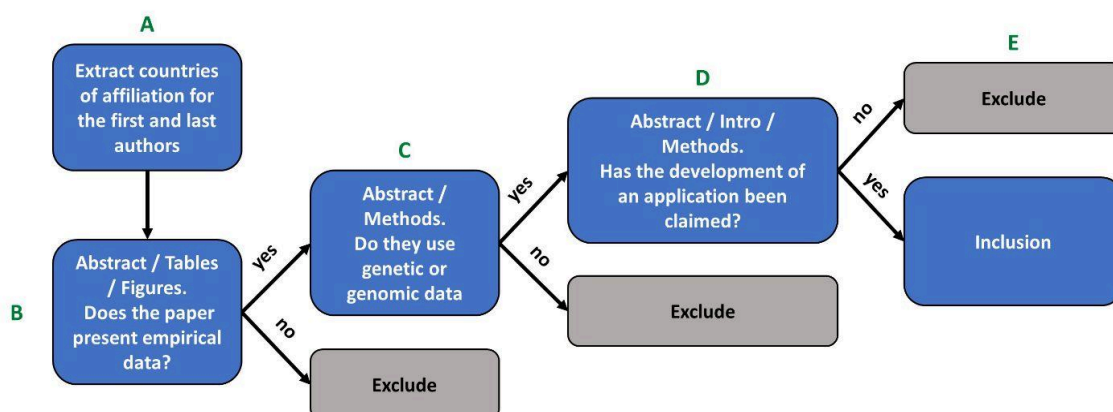


Figure 2. Inclusion flowchart

4.2. STEP 2 - Relevant manuscript screening for characterization of the potential genetic/genomic application and stakeholder identification

The articles that pass validation in STEP 1 will be subjected to a detailed assessment that requires the screening and careful reading of all sections.

Note that this screening focuses not only on the potential genetic/genomic application, but also on the stakeholder participation.

Stakeholders have diverse perspectives, backgrounds, and interests. We propose the following steps for the stakeholder screening.

Firstly, check if the stakeholders were mentioned in the authors' list, introduction, material & methods, discussion and/or acknowledgement. If they were mentioned, it means they were involved in research. In other parts of the manuscript, we leave it to the reviewer to decide if the description clearly mentions their involvement.

Then, you need to answer the following questions, which indicate their involvement:

- 1) What is the stakeholders role or level of involvement in the research project?
- 2) In which phase(s) of the study or research have stakeholders been involved
- 3) What type(s) of stakeholders have been considered in the research?

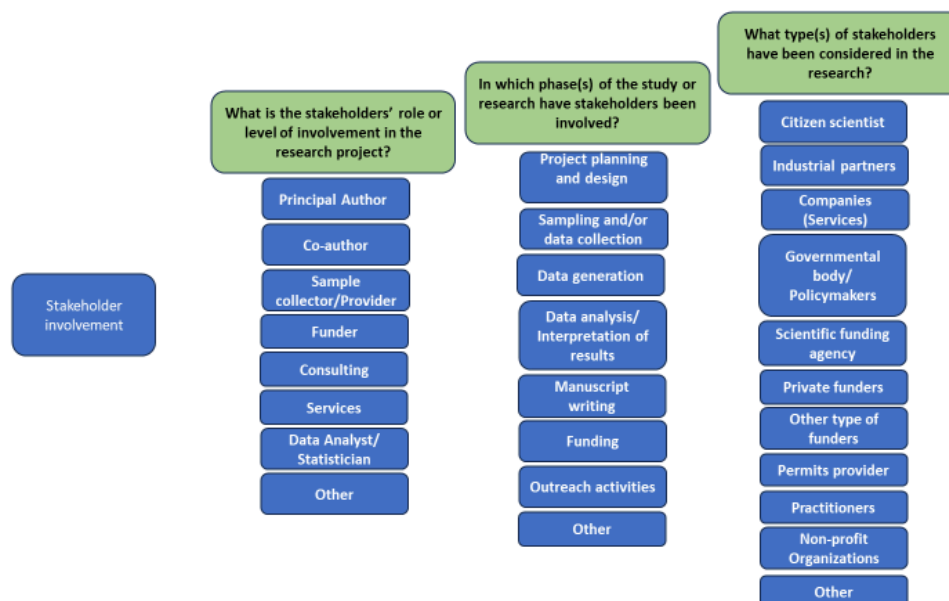


Figure 3. Stakeholders mapping

4.3. STEP 3 - Reanalysis of relevant articles to introduce redundancy in the Literature Review

After STEP 1 and STEP 2, which are part of the same temporal stage, articles that were deemed relevant for analysis will be redistributed by reviewers, to allow a minimum of a second assessment for each paper. This aims at identifying heterogeneity of assessments and increasing the robustness of the analysis. Temporally, STEP 3 will be done in a follow-up stage, after completion of STEPS 1 and 2.

5. Instructions for the Literature Review

5.1 - Description and examples of relevant content.

To integrate the Gap Analysis, it is essential for the article to provide a response to the task outlined in the Aims and Strategy section. To achieve this objective, the articles should incorporate one or more of the following elements:

i. From Basic to Applied Science

- Includes the use of genetics or genomics (with potential use of reference genomes) in conservation, bioeconomy or disease research.
- Identifies factors preventing the application of genomics in research projects in basic science and/or applications in conservation.
- Use of genetic or genomic data metrics to assess population/species status and trend. Which metrics were used and which taxa was analysed?
- Identifies gaps in the current tools applied in conservation genetics and genomics research, especially in the fields of conservation, bioeconomy and disease.
- Identifies barriers to the application of genomics in practical applications
- Determines ways to make genomic analysis accessible to conservation scientists and practitioners.

iii) From Applied Science to Stakeholders

- Characterises categories of stakeholders interest in conservation practices.
- Identifies type of collaboration between genomic scientists and key stakeholders, conservation practitioners and citizen scientists.
- Demonstrates the importance of the application of genomics in biodiversity management strategies.

5.2 - How to extract relevant information

Essential information from all research articles will be retrieved. This includes PubMed reference number, DOI, author names, country of affiliation, title of the publication, publication year and journal.

The data will be gathered using a [Google Spreadsheet](#). The reviewer should fill the spreadsheet available online. Make sure to thoroughly review the instructions provided in this SOP for each field to align your submitted data as closely as possible with the given guidance.

5.3 - Detailed instructions to fill the research article data

5.3.1. Pre-Filled sections (grey section of the spreadsheet)

A. PMID. PubMed reference number.

B. DOI. This field should record the unique alphanumeric string assigned to the research article, to provide a persistent link to its location on the internet.

C. Journal. This field should record the name of the scientific publication.

D. Title. This field should record the title of the research article.

E. Year. This field should include the year of publication.

F. First Author (last name, first name). Enter the name of the first author of the research article.

G. Country of affiliation. Multiple choice. The country of affiliation of a researcher refers to the country with which the first author is primarily associated or affiliated. It should be the first affiliation. All countries can be included. **This column must be completed by the reviewer.**

H. Last author (last name, first name). Enter the name of the last author of the research article.

I. Country of affiliation. Multiple choice. The country of affiliation of a researcher refers to the country with which the last author is primarily associated or affiliated. It should be the first affiliation. All countries can be included. **This column must be completed by the reviewer.**

J. File name in the database. Name of the PDF file in the database folder in Sync.

5.3.3. The following sections must be filled by the reviewer.

K. Empirical Genetic data or Genomic Data? Yes/no question. This field indicates whether empirical genetic or genomic data was employed to investigate or address the research question. It provides information on whether genetic data was utilised in the study to analyse and gain insights related to the research question.

L. Claim the development of an application? Yes/no question. The research paper mentions either the development and utilization of a genetic or genomic application or includes it as one of its objectives.

Examples:

- 10.1111/1365-2664.14540
- 10.1126/science.ade3984
- 10.1093/icesjms/fsac223
- 10.1111/j.1365-294X.2004.02274.x
- 10.1016/j.ygeno.2021.08.010
- 10.1111/j.1469-8137.2007.02175.x
- 10.1111/j.1365-294X.2008.03794.x

Note: If the answer is No to any of the fields K and L, do not extract any further information from the research article.

M. Article motivation: conservation, bioeconomy, disease, and other. Multiple choice. The motivation of the article should focus on conservation, bioeconomy, and disease, exploring the interplay and potential connections between these three important areas.

Conservation refers to the protection, preservation, and sustainable management of ecosystems and biodiversity. It involves efforts to safeguard species and ecosystems from threats such as habitat destruction, pollution, climate change, and overexploitation.

Bioeconomy refers to the utilisation of biological resources, such as agricultural crops, forestry products, and marine resources, for sustainable economic development. It involves the application of genomic approaches to utilise biological resources efficiently and promote sustainable practices.

Disease, in this context, can refer to various aspects, including infectious diseases, emerging diseases, or diseases affecting species. The article should promote the understanding of the relationships between disease dynamics, biodiversity, and ecosystem health, crucial for the effective conservation and sustainable bioeconomic practices.

N. Other motivation. If the answer in the previous field was 'other', please write the motivation in this field. If more than one, separate them with a comma.

O. Species name. The species epithet should be written in this field. In case that more than one species is targeted, list the species names separated by a backslash: *Panthera leo/Canis lupus*.

P. Taxonomic group. Multiple choice. The taxonomic Order into which the Family is placed or the monophyletic group to which the family or Genus belongs. This should correspond to the taxonomy as represented in the NCBI Taxonomy Database.

Q. Countries of origin of the biological samples. Multiple choice. This field should record the countries of origin of the samples included in the research study.

R. Reference Genome. Yes/no question. This field indicates whether a reference genome was employed to address the research question.

S. Reference Genome from the target species. Yes/no question. This field indicates whether a reference genome belonging to the target species was used in the research paper. The name of the species should also be mentioned.

T. Chromosome level-assembly? Yes/no question. This field indicates whether the reference genome used in the research project is at chromosome level genome assembly.

U. High-throughput genetic data? Yes/no question. High-throughput genetic data can be defined as large-scale genomic sequencing or genotyping techniques that enable the examination of more than 100 to millions of genetic variants across the genome. Several types of data that could be considered high-throughput genetic data, include:

- Whole-genome sequencing (WGS),
- Pool Sequencing
- Exome sequencing,
- Transcriptome sequencing,
- RAD sequencing
- Genotyping array,
- Targeted capture.

V. Type of Genomic Data. Multiple choice. Indicate the type of genomic data used in the research project.

W. Other. Specify if other type of genomic data were used. If more than one use a comma to separate the different kinds of data.

X. Low-throughput genetic data? Yes/no question. Low-throughput genetic data can be defined as lower-scale genomic sequencing or genotyping techniques that enable the examination of less than 100 variants across the genome. Several types of data that could be considered low-throughput genetic data, include:

- Short-tandem repeats (STRs), e.g. microsatellites, are regions of DNA where short sequences of nucleotides are repeated. These repeats are typically 2 to 6 base pairs in length and are consecutively repeated. The number of repeats can vary among individuals, and STRs are commonly used in DNA profiling and forensic analysis.
- Single nucleotide polymorphism (SNPs)
- Sanger sequencing, is a method for determining the sequence of nucleotide bases in a DNA molecule.
- Barcoding.

Y. Type of Genetic Data. Multiple choice. Indicate the type of genetic data used in the research project.

Z. Other. Specify if other type of genetic data were used. If more than one use a comma to separate the different kinds of data.

AA. Nuclear DNA. Yes/no question. This field specifies whether the genetic markers originate from nuclear DNA.

AB. mtDNA. Yes/no question. This field specifies whether the genetic markers originate from mtDNA.

AC. cpDNA. Yes/no question. This field specifies whether the genetic markers originate from chloroplasts.

AD. Bacteria/Virus DNA. Yes/no question. This field specifies whether the genetic markers originate from bacteria or viruses.

AE. Which genetic metrics are employed for the characterization of species and/or populations? Multiple choice. Genetic metrics are quantitative measures that are employed to assess and describe the genetic diversity, structure, and relationships within species and populations. These summary statistics could include metrics to describe the genetic variation within species/populations:

- Nucleotide Diversity
- Allele Frequency
- Heterozygosity
- Fixation Index (FST)
- Tajima's D
- Wright's F-statistics (FIS, FST, and FIT)
- Pairwise Genetic Distance
- Linkage Disequilibrium (LD)
- Principal Component Analysis (PCA)
- Neighbour-Joining Tree
- Demographic History Models
- Effective Population Size (Ne)
- Runs of Homozygosity (ROH)
- Structure Assignment
- Isolation by distance
- Single locus phylogeny
- Multilocus phylogeny
- Networks

AF. Other. If any other metric was used and it is not included in the previous field, list them in this field separated by a comma.

AG. Does an application arise from the research project? Yes/no question. This field indicates whether an application within the domains of conservation, bioeconomy and disease was produced as a result of the research project. An application means that the results from the work can be directly used by practitioners or non-scientists to address the particular domain in the model species, without additional Research and Development.

Examples:

- 10.1111/1365-2664.14540
- 10.1126/science.ade3984
- 10.1093/icesjms/fsac223
- 10.1111/mec.12227

AH. Immediate Application. Yes/no question. This field indicates that the work has developed a toolkit that is being used by practitioners or non-scientists. For example, this means that the paper itself describes the application of the toolkit by practitioners or non-scientists.

Examples:

- 10.1111/1365-2664.14540
- 10.1093/icesjms/fsac223

AI. Future Application. Yes/no question. This field indicates whether the toolkit has the potential of being applied in the field by practitioners or non-scientists in the future, but the toolkit has not yet been applied in the field. For example, the paper generates knowledge or a toolkit that can be directly used by non-scientists or practitioners in the future but its application is not described in the article.

Examples:

- 10.1126/science.ade3984
- 10.1111/mec.12227

AJ. Are the relevant stakeholders involved? Yes/no question. In the context of conservation, bioeconomy and disease, various stakeholders play important roles in preserving and managing natural resources, ecosystems, and biodiversity.

AK. What is the stakeholders role or level of involvement in the research project?

Dropdown menu with the following items:

- 1) Principal Author
- 2) Co-author
- 3) Sample Collector/Provider
- 4) Permits provider
- 5) Funder
- 6) Consulting
- 7) Services (sequencing centre; bioinformatics; etc)
- 8) Data Analyst/Statistician
- 9) Other (please specify in the following column)

AL. Specify if others. Use the comma to separate multiple items.

AM. In which phase(s) of the study or research have stakeholders been involved?

Dropdown menu with the following items:

- 1) Project planning and design
- 2) Sampling and/or data collection (permits; etc)
- 3) Data generation
- 4) Data analysis/Interpretation of results
- 5) Manuscript writing
- 6) Funding acquisition
- 7) Outreach activities (workshops; etc)
- 8) Other (please specify in the following column)

AN. Specify if others. Use the comma to separate multiple items.

AO. What type(s) of stakeholders have been considered in the research? Dropdown menu with the following items:

- 1) Industry Partners/SMEs
- 2) Companies (Lab platform for services)
- 3) Governmental body/ Policymakers (International)
- 4) Government body/Policymakers (National)
- 5) Government body/Policymakers (Regional/Local)
- 6) Scientific Funding Agency (International)
- 7) Scientific Funding Agency (National)
- 8) Scientific Funding Agency (Regional/Local)
- 9) Private funders (International),
- 10) Private funders (National)
- 11) Private funders (Regional/Local)
- 12) Other type of funders (charity; parks; not clear; etc)
- 13) Practitioners (National parks; Zoos; Botanical gardens; Museums; etc)
- 14) Practitioners (Non-profit organisations)
- 15) Permits provider (International)
- 16) Permits provider (National)
- 17) Permits provider (Regional/Local)
- 18) Citizen scientist/Community Representatives ##(volunteers, hobbyists, amateurs, naturalists, hunters, fishers, wildlife observers, landowners, artists)
- 19) Other (please specify in the following column)

AP. Specify if others. Use the comma to separate multiple items.

AQ. Do any ethical issues mentioned in the article? Ethical issues may arise from the use of genomic data in research. The article may provide a comprehensive discussion of the specific ethical considerations and frameworks relevant to the research project and its implications for biodiversity protection:

- 1) Scientific approach
- 2) Practicioners
- 3) Policy
- 4) Local community/indigineous people

AR. Specify if others. Use the comma to separate multiple items.

6. Definition of Gap Analysis Team

The work associated with the Gap analysis leading to the development of a report (and associated outputs) will be performed by the **Gap Analysis Team**. The Gap Analysis team members are expected to make meaningful contributions to ensure the success of the analysis. The active engagement of team members during the process can significantly impact the quality and outcome of the gap analysis, allowing the achievement of the analysis's objectives.

Minimal Expected contribution

- **Peer Review Participation:** Analyse and review research papers (expected contribution: **analysis of 200 papers**) contributing to enhance the quality and rigour of the analysis. ***Note that the number of papers to be reviewed can change depending on the number of elements in the Gap Analysis Team.***

Gap Analysis Core Team

- **Participation in Meetings:** Attend scheduled meetings and actively engage in discussions by sharing your insights, perspectives, and ideas.
- **Gap Identification:** Contribute to the identification of gaps by analysing existing research, literature, and data to pinpoint areas where knowledge or solutions are lacking.
- **Solutions and Recommendations:** Contribute with suggestions for potential solutions or strategies to address the identified gaps. Your input can help shape the direction of the analysis.

Nature of the Collaboration:

All members of the Gap Analysis Team are regarded as active collaborators and co-authors for the forthcoming Gap Analysis publication. It is understood that each team member is aligned with and supportive of the Open Access Data Sharing policy that has been established by BGE (insert link once available). This policy adheres steadfastly to the principles of FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics), as stipulated by the European Union.

It is significant to note that the flexibility to include additional members remains open throughout the development of the Gap Analysis. This inclusive approach allows for the integration of new collaborators as the study evolves, thereby enriching the analysis process and outcomes.